LECTURE 6

LAST TIME

LECTURE 6: SUMMARY

Hedging forwards

Risk management

MGRM

THIS TIME

LECTURE 7 OUTLINE

Introduction to options

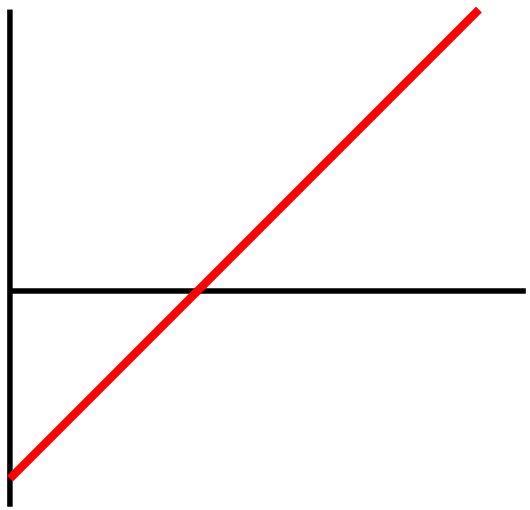
No-arbitrage bounds on options prices

TYPES OF DERIVATIVES

TYPES OF DERIVATIVES

linear

e.g. forwards, futures, swaps



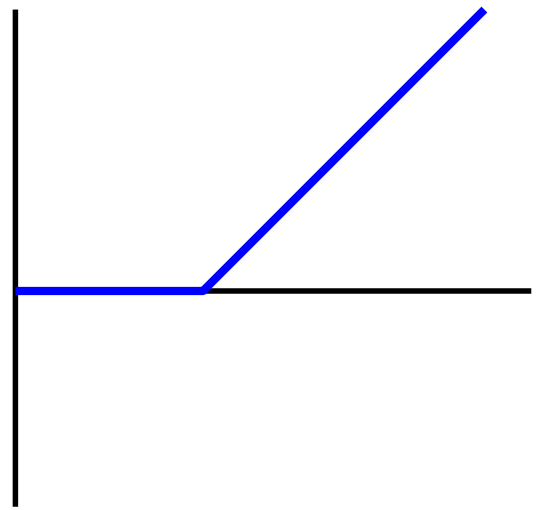
easy (or easier)

priced w/o models

static hedge

non-linear

e.g. options



difficult

require models to price

dynamic hedge

but model-free NA bounds

OPTION DEFINITIONS

DEFINITIONS: EUROPEAN CALLS AND PUTS

A European call option is a contract giving

the right to buy an underlying asset

at a pre-specified strike price

at pre-specified maturity

A European put option is a contract giving

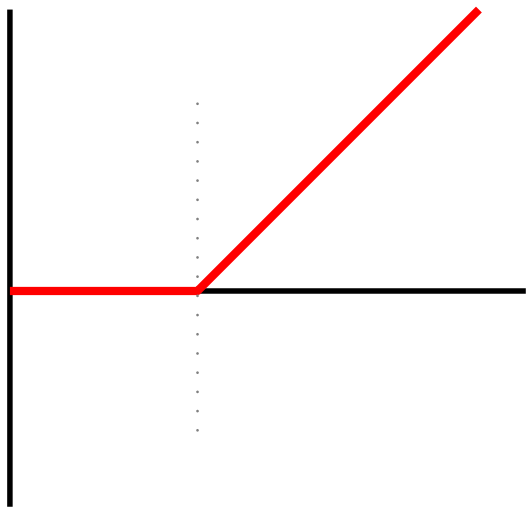
the right to sell an underlying asset

at a pre-specified strike price

at pre-specified maturity

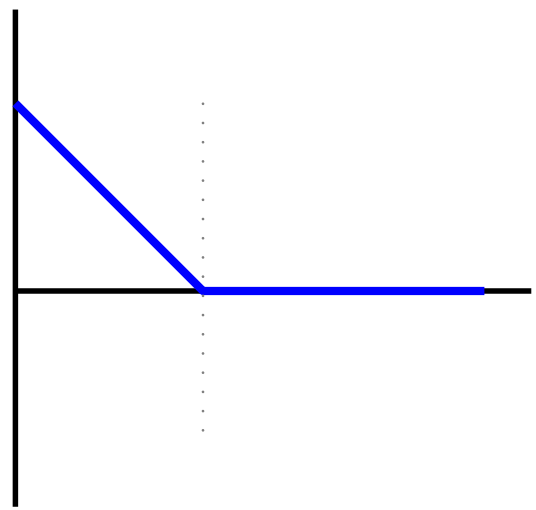
CALL AND PUT PAYOFF AT EXERCISE

call



exercise if strike
below market

put



exercise if strike
above market

DEFINITIONS: AMERICAN CALLS AND PUTS

An American call option is a contract giving

the right to buy an underlying asset

at a pre-specified strike price

when exercised at any time before pre-specified maturity

An American put option is a contract giving

the right to sell an underlying asset

at a pre-specified strike price

when exercised at any time before a pre-specified maturity

OPTIONS PAYOFFS

An option's payoff is defined by optionality

The holder of the option chooses the greater of two payoffs

For a call with strike K and maturity T on a stock S

$$c_T = \max\{0, S_T - K\} \equiv \big(S_T - K\big)\mathbb{1}_{\{S_T \geq K\}} =: \big(S_T - K\big)^+$$

For a put with strike K and maturity T on a stock S

$$p_T = \max\{0, K - S_T\} \equiv \big(K - S_T\big)\mathbb{1}_{\{S_T \leq K\}} =: \big(K - S_T\big)^+$$

where for an event (a set) A, the indicator function $\mathbb{1}$ is

$$\mathbb{1}_A(\omega) := \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{otherwise} \end{cases}$$

OPTION ASYMMETRY

Optionality makes payoff asymmetric and non-linear

The call's upside—above the strike—linear and increasing

The call's downside—below the stike—zero (but premium sunk)

Convexity at the kink

HEDGING WITH OPTIONS

HEDGING COCOA WITH OPTIONS

Chocolate producer is short cocoa

Wishes to hedge risk over price to buy cocoa

Why not agree to a maximum price to pay? I.e. hedge with a call

Call his chocolate revenue if R, cocoa price S, strike K

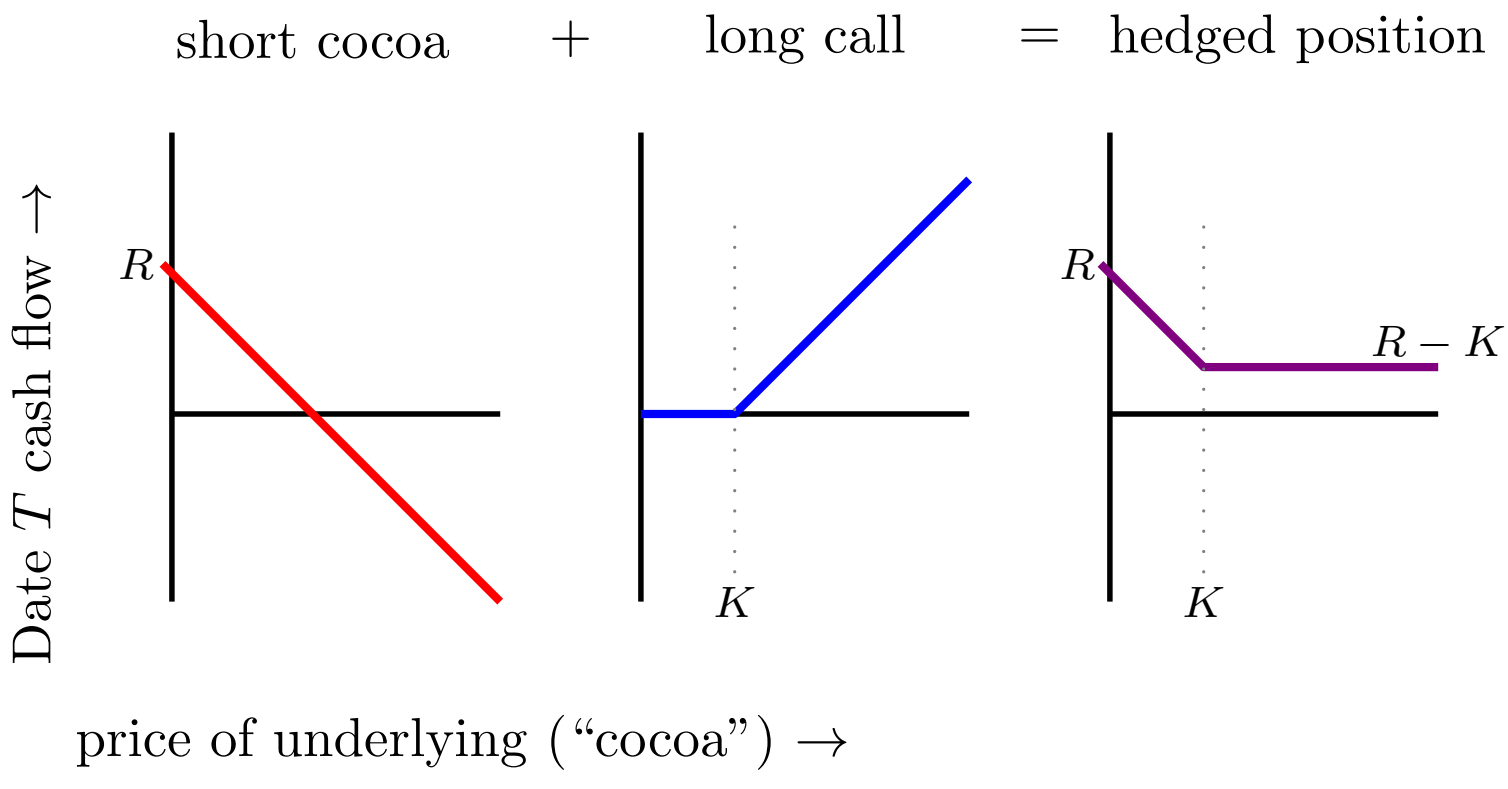
If the call is in the money, pay K

$$\text{if } S_T \geq K \text{ then } \mathrm{CF}_T = R - K$$

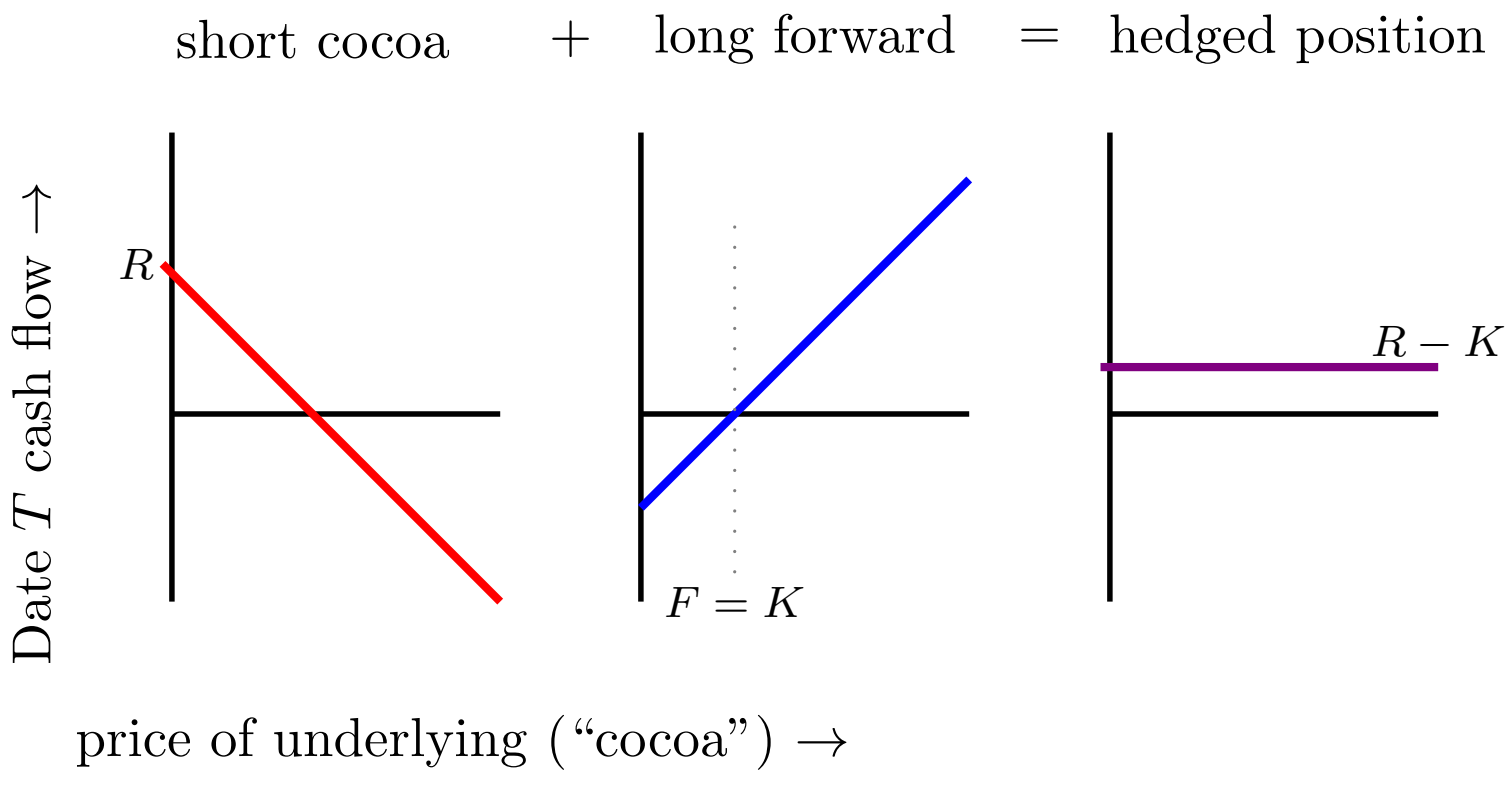
If the call is out of the money, all the better, pay $S_T < K$

$$\text{if } S_T < K \text{ then } \mathrm{CF}_T = R - S_T$$

HEDGING WITH A CALL

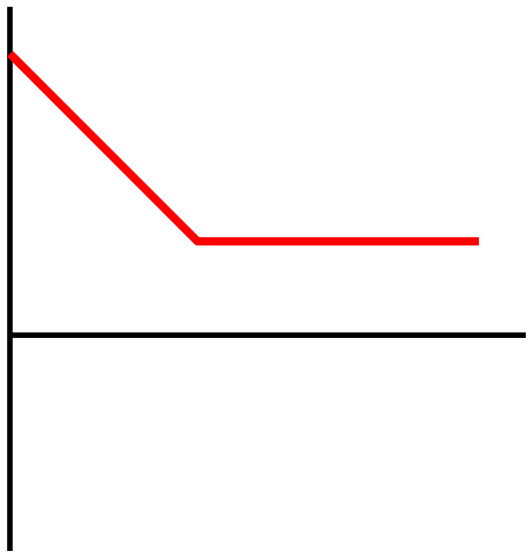


RECALL HEDGING WITH A FORWARD



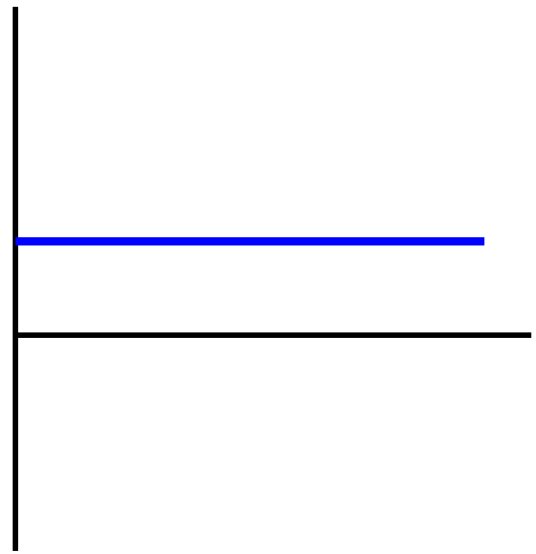
PAYOFF FROM CALL HEDGING DOMINANTES

call hedge



$\max\{R - S_T, R - K\}$

forward hedge



$R - K$

Why would you ever hedge with a forward?

CHERRY-PICKING RIGHT/INSURANCE VALUE

Holding option comes with "cherry-picking right"

- Right to exercise only if advantageous
- But pay a premium upfront

Cherry-picking right is the insurance value of the option

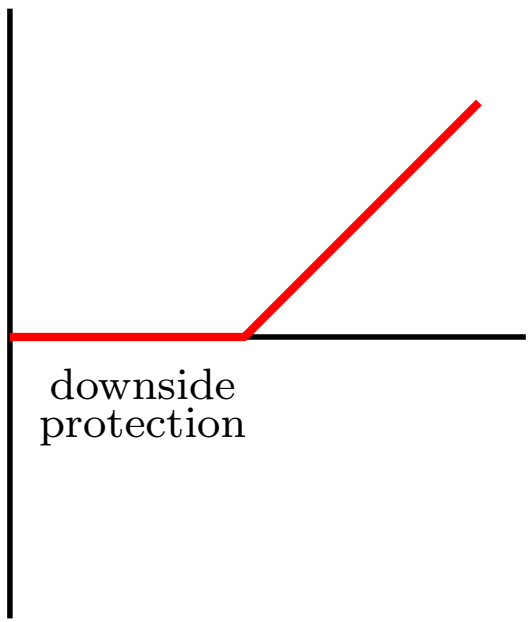
Contrast the insurance value with the speculative value

- Speculative value refers to upside value call inherits
- Speculative value is already in underlying and forward
- Calls differ due to downside protection

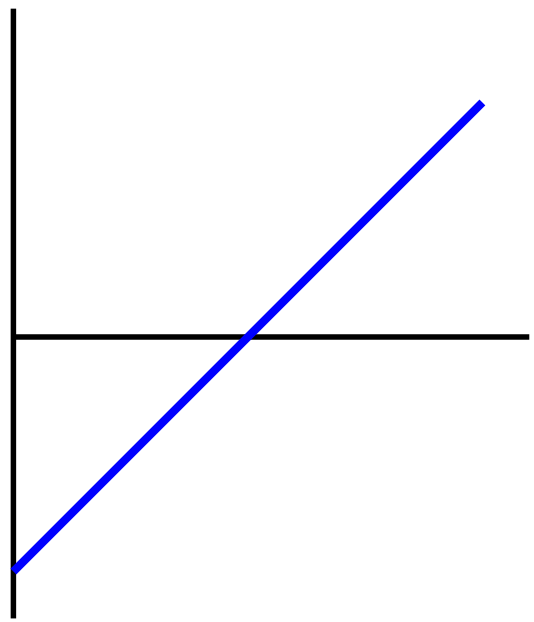
OPTION PRICE DECOMPOSITIONS

INSURANCE VALUE OF CALL VS. FORWARD

call



forward



INSURANCE VALUE DECOMPOSITION: CALL

Recall formula for value of forward with forward price $F_t(T) = K$

$$V_t = e^{-y(T-t)} S_t - e^{-r(T-t)} K$$

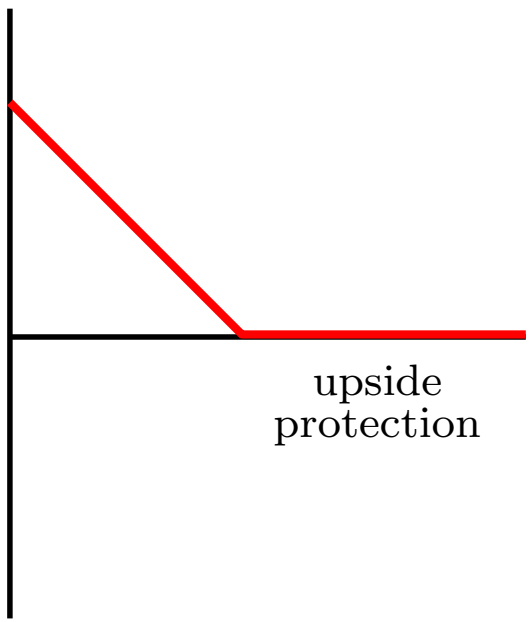
Use this to decompose call price

$$\begin{aligned} c(t, S_t; K, T) &= \text{ value of forward } + \text{ insurance value} \\ &= \left[e^{-y(T-t)} S_t - e^{-r(T-t)} K \right] + \text{ insurance value} \end{aligned}$$

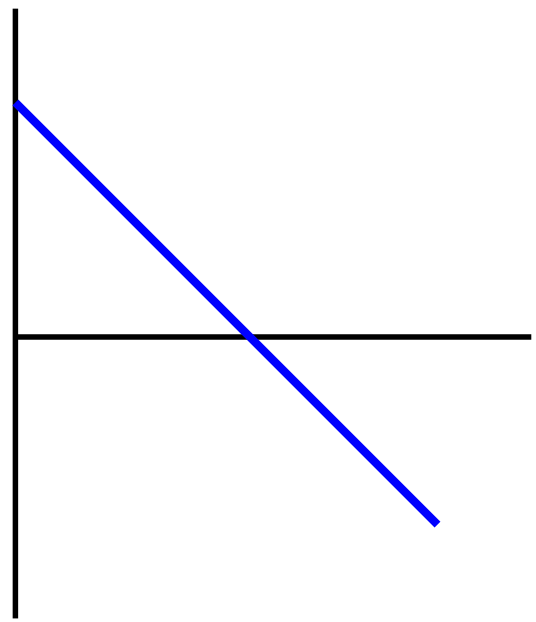
A call is a forward (forward price K) plus option to default

INSURANCE VALUE OF PUT VS. FORWARD

put



short forward



INSURANCE VALUE DECOMPOSITION: PUT

Use formula for value of short forward to decompose put price

$$\begin{aligned} p(t, S_t; K, T) &= \text{ value of short forward } + \text{ insurance value} \\ &= \left[e^{-r(T-t)} K - e^{-y(T-t)} S_t \right] + \text{ insurance value} \end{aligned}$$

A put is a short forward (forward price K) plus option to default

For put, insurance value is on upside of short position

INTRINSIC VALUE

Another decomposition is based on intrinsic value and time value

Intrinsic value is the value if exercised today

$$\begin{aligned} \text{intrinsic value of call} &= (S_t - K)^+ \\ \text{intrinsic value of put} &= (K - S_t)^+ \end{aligned}$$

Note: for traders sometimes forward is underlying; then

$$\begin{aligned} \text{intrinsic value of call} &= \big(F_t(T) - K\big)^+ \\ \text{intrinsic value of put} &= \big(K - F_t(T)\big)^+ \end{aligned}$$

VOCABULARY: _____ THE MONEY

Option is in the money if intrinsic value is positive

Option is out of the money if intrinsic value is zero

Option is at the money (ATM) if $S_t \approx K$

TIME VALUE DECOMPOSITION: CALL

The time value of the call is the value above the intrinsic value

$$c(t, S_t; K, T) = \text{ intrinsic value of call } + \text{ time value}$$
$$= \left(S_t - K\right)^+ + \text{ time value}$$

Time value is the sum of the values of two effects

Deferred payment: pay K later, get $K(e^{r(T-t)} - 1)$ in interest

Potentiality/Cherry-picking: more t.t.m., more vol till payoff

TIME VALUE DECOMPOSITION EXERCISE

A call with maturity 1 year and strike 120 has price 2

The underlying stock has price 100

The interest rate is 0

What is the intrinsic value

- the deferred payment value?
- the potentiality value?
- the time value?

VALUE DECOMPOSITION SOLUTION

$S_t < K$ so the intrinsic value is zero

r is zero, so the deferred payment value is zero

Thus the potentiality value is the time value is the price, 2

TIME VALUE DECOMPOSITION: PUT

The time value of the put is the value above the intrinsic value

$$\begin{aligned} p(t, S_t; K, T) &= \text{ intrinsic value of put } + \text{ time value} \\ &= \left(K - S_t\right)^+ + \text{ time value} \end{aligned}$$

Time value is the sum of the values of two effects

Deferred payment: get K later, lose $K(e^{r(T-t)} - 1)$ interest

Potentiality/Cherry-picking: more t.t.m., more vol till payoff

Note for puts, time value can be negative

since deferred payment value is negative

PUT-CALL PARITY

PUT-CALL PARITY

A call is a long forward plus downside protection

A put is a short forward plus upside protection

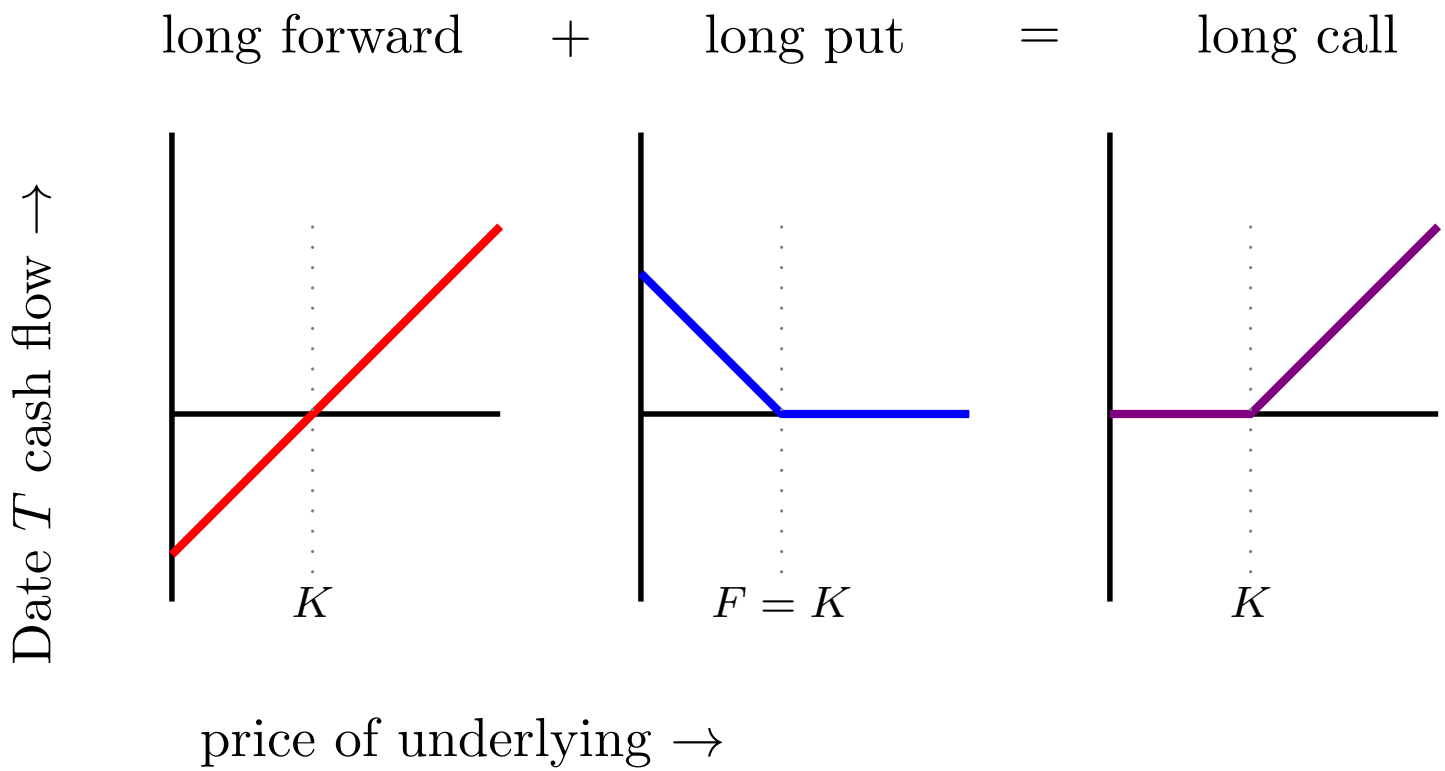
Put call parity is the result that as long as there is NA

value of call's downside protection = value of put's upside protection

or

value of call = value of forward with price K + value of put

PUT-CALL PARITY FIGURE



PUT-CALL PARITY SPECULATION EXERCISE

Suppose you are bearish on a stock and want to speculate

But, for institutional reasons, you cannot borrow it to short

e.g. in August 2008 the SEC outlawed naked short sales

Put-call parity says

$$-S_t = -c_t + p_t - e^{-r(T-t)}K$$

you can create a synthetic short position

write a call, buy a put, borrow PV(K) dollars

PUT-CALL PARITY PROOF

Could suppose strict inequalities and show contradiction

Instead, use the Law of One Price

Payoff of long call plus short put

$$\begin{aligned} c_T - p_T &= \max\{0, S_T - K\} - \max\{0, K - S_T\} \\ &= S_T - K \end{aligned}$$

This is the forward or, if no dividends, long stock, short K bonds

NO-ARBITRAGE BOUNDS

MODEL-FREE BOUNDS

Option pricing formulae need assumptions on stock dynamics

But general restrictions (inequalities) on option prices do exist

AMERICANS ARE BETTER THAN EUROPEANS

$$C(t, S; T, K) \geq c(t, S; T, K)$$

$$P(t, S; T, K) \geq p(t, S; T, K)$$

OPTIONS PRICES ARE POSITIVE

$$c \geq 0$$

$$p \geq 0$$

UPPER BOUNDS

$$C \leq S$$

$$P \leq Ke^{-r(T-t)}$$

RECAP

LECTURE 7: RECAP

Introduction to calls and puts

Hedging with options vs. hedging with forwards

Option value decompositions

- value of forward + insurance value
- intrinsic value + time value

Put-call parity

FIN